

URBAN CLIMATE INTELLIGENCE PLATFORM

Mumbai's ward-level heat vulnerability, turned into decisions you can act on.

Tells Mumbai's city planners which of the 24 BMC wards to cool first, why, with what intervention, and where a fixed budget goes.

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GRADE	11 (IBDP Year 1)
SELECTED DOMAIN	PLANET
FOCUS AREA	Urban heat resilience / climate adaptation decision-support

● What's the problem?

Mumbai heats unevenly. Slums, bare land, and concrete trap the most heat, but the city has no ward-by-ward, evidence-based way to decide where cooling money goes first. Existing plans set city-wide goals and stop there.

● Who's affected?

Slum households in heat-trapping housing, the elderly, and anyone far from a hospital. The people most at risk live exactly where the data is thinnest.

● Why it matters

Heat kills more people in India than any other weather hazard. Ahmedabad's Heat Action Plan saves an estimated 1,190 lives a year by targeting action locally (Knowlton et al. 2014). Mumbai's own climate plan names heat as a priority but never gets past city-level strategy to a method for picking which ward first.

● Where it happens

Every dense tropical city faces this. UCIP starts with Mumbai's 24 BMC wards, where IIT Bombay's remote-sensing work already shows sharp heat-island effects over informal housing (Mehrotra, Bardhan and Ramamritham 2018).

• What the data shows

- A live satellite pipeline over Mumbai (Google Earth Engine, 12 Landsat 8/9 dry-season scenes) puts mean land-surface temperature at 33.6°C and mean NDVI at 0.305, computed directly, not looked up.
- WorldPop's 2020 age-sex rasters (the most recent year for India) give elderly population per ward; OpenStreetMap gives hospital locations; Datameet's mapped slum clusters stand in for a slum index.
- Tree-canopy cooling is nonlinear: below ~40% canopy, daytime cooling is negligible; 40 to 80% delivers about 1°C (Ziter et al. 2019). Cool roofs cut peak temperature roughly 0.6K per +0.1 albedo, the conservative end of the published range (Santamouris 2014).

• What's already out there

Mumbai Climate Action Plan 2022	Sets city-wide goals, no per-ward ranking method
RAND / Azhar et al. 2017 India HVI	District-level across 640 districts, too coarse for one city
IIT Bombay Mumbai UHI studies	Rigorous heat maps, diagnostic only
C40 Urban Cooling Toolbox	Real decision-support, but generic, no Mumbai data underneath
Ahmedabad Heat Action Plan	Warns when heat strikes, doesn't say where to build

• The gap

No existing tool does all five things a planner actually needs: ward-level ranking, per-factor explanations, an intervention engine, an ecological check, and a budget layer. Most also overstate their own certainty and stay quiet about where their data is a proxy.

- **The idea**

UCIP scores a Heat Vulnerability Index for each of Mumbai's 24 wards from satellite and demographic data, weighted by the peer-reviewed Reid et al. 2009 method rather than guesswork. Every ward gets a transparent per-factor breakdown, and the ranking is stress-tested by perturbing the weights $\pm 20\%$ to confirm it holds.

- **How it solves the problem**

That index feeds a rule-based recommendation engine where every suggestion carries a citation. An ecological filter checks first: native trees where restoration actually makes sense (Bastin et al. 2019), cool roofs instead wherever planting would backfire over native grassland (Veldman et al. 2019). It's not another "plant trees everywhere" map.

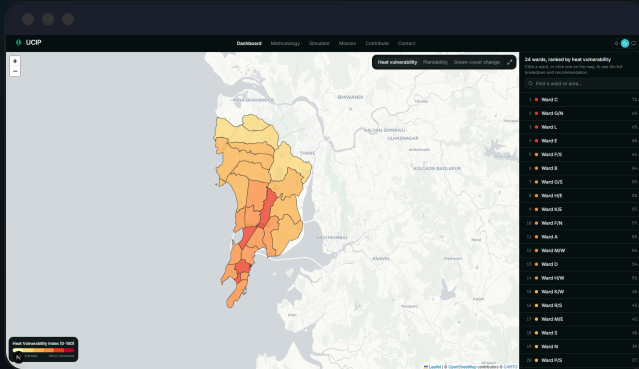
- **Who benefits**

BMC planners get a defensible priority list instead of a guess

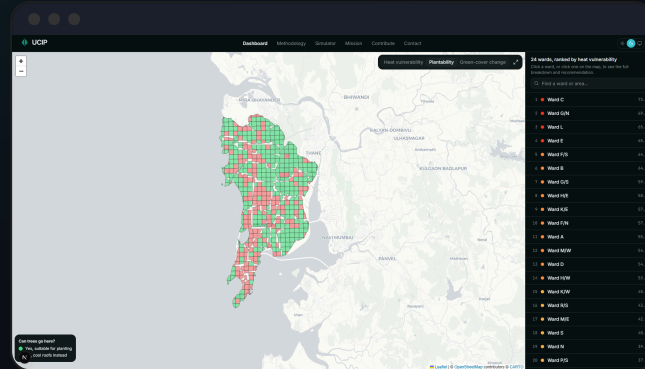
Vulnerable residents get cooling investment where it's needed most

Researchers and NGOs get open methodology, open data, stated limits

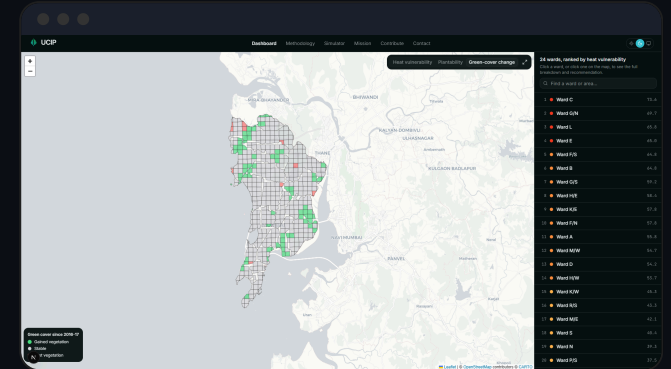
05. PROTOTYPE / SOLUTION DESIGN



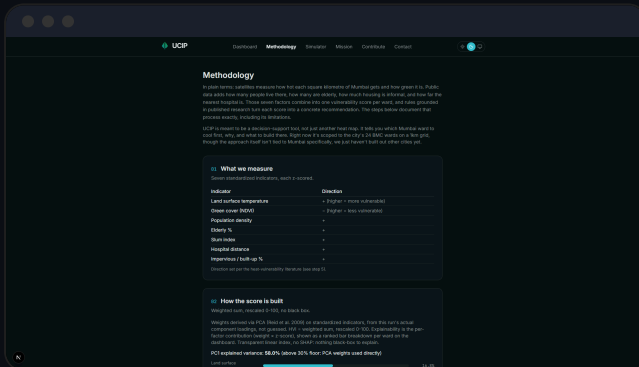
The 24-ward HVI ranking, plus per-ward factor bars and the recommended intervention



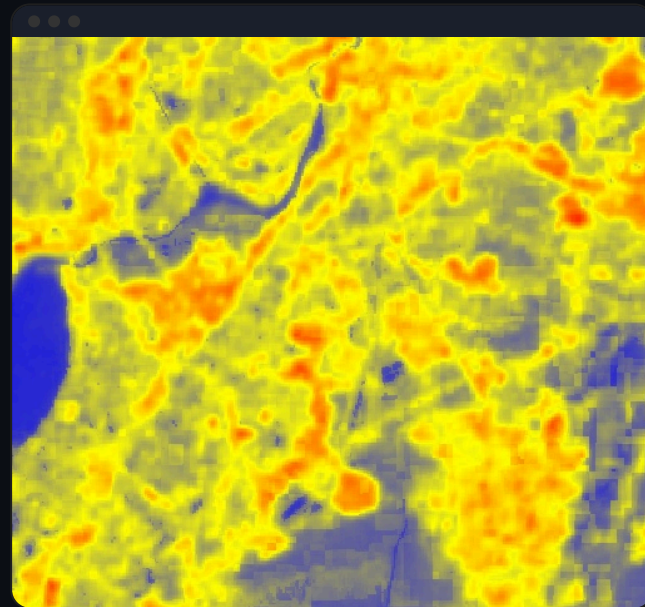
The plantability filter: green cells suit afforestation, red cells route to cool roofs instead



Green-cover change: gained, stable, or lost NDVI per cell



The methodology page: real computed weights, sensitivity chart, citations, stated limits



Pipeline evidence: raw GEE land-surface-temperature composite



Pipeline evidence: raw GEE NDVI composite

Google Earth Engine (Landsat 8/9, WorldPop, WorldCover) → Python (geopandas, scikit-learn PCA) → Supabase + PostGIS → Next.js + Leaflet, deployed on Vercel.

- **Expected impact**

Mumbai already knows it's getting hotter. UCIP turns that diagnosis into a decision: which ward first, why, what to build, at what cost. The same method works for any Indian city on the same free data (Landsat, WorldPop, OSM, GHSL). Every limitation is stated up front: proxy data flagged, land-surface temperature never confused with air temperature, coefficients marked as borrowed from other cities, not Mumbai-measured.

- **What's next**

Finer grid (1km to 500m) and census-linked demographics once accessible

NDVI change tracking to confirm interventions actually greened a ward

What-if simulator: slide canopy % or cool-roof fraction, watch LST move (shipped)

Budget optimizer: maximize vulnerability reduction per rupee spent (stretch)

Replication kit: same config, different city (Pune, Ahmedabad, Delhi)